

Unified Smart Laboratory Ecosystem

Laboratory Technician Questionnaire

Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Research Study: IoT-Based Smart Laboratory Systems in Higher Education

INSTRUCTIONS TO RESPONDENTS

1. This questionnaire is for research purposes only. All responses are strictly confidential.
2. Please answer all questions as accurately and honestly as possible.
3. For Likert scale questions, circle the number that best represents your opinion.
4. For multiple choice questions, tick (✓) all options that apply unless stated otherwise.
5. For open-ended questions, write your response in the lines provided.
6. The questionnaire should take approximately 20–25 minutes to complete.

By completing this form, you consent to your responses being used for academic research.

SECTION A: Respondent Information

This section is used for demographic analysis only and will not be used to identify individual respondents.

Q1. What is your current role in the laboratory?

- | | |
|---|---|
| ☐ Laboratory Technician | ☐ Research Assistant |
| ☐ Senior Laboratory Technician | ☐ Other (specify): _____ |
| ☐ Laboratory Superintendent | |

Q2. Which department/laboratory do you work in? (Tick all that apply)

- | | |
|--|--|
| ☐ Physics Laboratory | ☐ Computer Laboratory |
| ☐ Chemistry Laboratory | ☐ Biology Laboratory |
| ☐ Engineering Laboratory | ☐ Other: _____ |
| ☐ Clinical/Medical Laboratory | |

Q3. How many years of experience do you have in laboratory management?

- | | | |
|---|---------------------------------------|---|
| ☐ Less than 1 year | ☐ 4 – 6 years | ☐ More than 10 years |
| ☐ 1 – 3 years | ☐ 7 – 10 years | |

Q4. What is your highest level of education?

- | | | |
|--------------------------------------|--|---------------------------------------|
| ☐ Certificate | ☐ Bachelor's Degree | ☐ Other: _____ |
| ☐ Diploma | ☐ Master's Degree | |

Q5. Have you previously used any digital or computer-based laboratory management system?

- | | | |
|------------------------------|-----------------------------|-----------------------------------|
| ☐ Yes | ☐ No | ☐ Not Sure |
|------------------------------|-----------------------------|-----------------------------------|

If yes, please explain:

SECTION B: Current Laboratory Practices

This section evaluates the current methods used to manage laboratory operations at your institution.

Q6. How is student attendance currently recorded in your laboratory? (Select all that apply)

- ☐ Physical sign-in register
- ☐ Digital system (please specify below)
- ☐ Lecturer/technician verbal roll call
- ☐ No formal record is kept
- ☐ Student identity card scan

Q7. How do you currently track laboratory equipment usage?

- ☐ Manual logbook
- ☐ Equipment is not formally tracked
- ☐ Spreadsheet (Excel/Google Sheets)
- ☐ Other: _____
- ☐ Dedicated software system

Q8. How are practical sessions currently documented?

- ☐ Students write reports in physical lab notebooks
- ☐ No formal documentation system
- ☐ Printed report templates submitted on paper
- ☐ Other: _____
- ☐ Digital reports submitted via email or portal

Q9. What are the most significant challenges you face in managing laboratory sessions?

Note: Be as specific as possible. Examples: missing equipment, student disputes, poor record keeping.

Q10. How would you rate the overall efficiency of your current laboratory management system?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q11. How frequently do issues arise due to poor or incomplete documentation of practical sessions?

- ☐ Never
- ☐ Sometimes (monthly)
- ☐ Very often (daily or almost daily)
- ☐ Rarely (once per semester)
- ☐ Often (weekly)

Q12. How are disputes about student attendance or practical completion currently resolved?

SECTION C: Awareness and Initial Perception of the Proposed USLE

The Unified Smart Laboratory Ecosystem (USLE) is a proposed integrated digital platform combining biometric authentication, IoT-based equipment tracking, digital lab notebooks, technician approval workflows, and analytics. Please respond based on this description.

Q13. Before reading the above description, were you aware of any similar smart laboratory management systems?

- ☐ Yes, I am aware of several such systems
- ☐ No, I was not aware of any
- ☐ Yes, I have heard of one or two
- ☐ I have used such a system before

Q14. After reviewing the USLE description, how would you rate its relevance to the needs of your laboratory?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q15. Which of the following proposed USLE features do you find most useful? (Tick all that apply)

- ☐ Biometric fingerprint authentication
- ☐ Group submission with individual verified copies
- ☐ RFID card fallback authentication
- ☐ Technician digital approval and signature
- ☐ Staff-issued time-limited QR code (emergency fallback)
- ☐ Analytics and performance dashboards
- ☐ IoT-based equipment tracking
- ☐ Cross-lab access for students
- ☐ Digital lab notebook
- ☐ Integration with Clinical/LIMS systems

Q16. What concerns or reservations do you have about implementing a system like USLE in your laboratory?

Q17. How strongly do you agree that a system like USLE would address the challenges you described in Section B?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

SECTION D: Authentication System Feedback

Q18. How reliable do you consider your current method of verifying student identity at the start of a practical session?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q19. Do you believe biometric fingerprint authentication would improve student accountability in your laboratory?

☐ Yes ☐ No ☐ Not Sure

If yes, please explain:

Q20. How practical do you think it is to enrol all students in a fingerprint recognition system at the start of each academic year?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q21. How useful would a technician-controlled, time-limited QR code be as a fallback when biometric authentication is unavailable?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q22. How concerned are you about the possibility of students bypassing or misusing the authentication system?

☐ Not concerned at all ☐ Moderately concerned ☐ Extremely concerned
☐ Slightly concerned ☐ Very concerned

Q23. Do you foresee any specific technical or operational challenges with implementing a multi-layer authentication system?

Q24. How important is it that the system can operate offline during network outages, with data synced later?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

SECTION E: Digital Lab Notebook and Reporting

Q25. How are student experimental observations and results currently recorded during practical sessions?

☐ Handwritten in physical lab notebooks ☐ No standard method — varies by student
☐ On printed lab sheet templates ☐ Other: _____
☐ Verbally recorded and written up later

Q26. How much do you think a structured digital lab notebook would improve the quality and completeness of student records?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q27. How practical do you consider it for students to use laptops or tablets to record data during laboratory experiments?

Note: Consider factors such as available space, power sockets, and type of practical.

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q28. Would you support a group submission model where one student submits a practical report on behalf of the group and the system automatically distributes verified individual copies to all group members?

☐ Yes ☐ No ☐ Not Sure

If yes, please explain:

Q29. How confident are you that a digital lab notebook would reduce cases of copied, forged, or plagiarised practical reports?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q30. What specific features should a digital lab notebook include to best meet the needs of your laboratory?

SECTION F: Technician Approval and Digital Signatures

Q31. How do you currently verify and confirm that a student has successfully completed a practical session?

Q32. How much would a formal technician digital approval step improve accountability in the submission process?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q33. How valuable do you consider a cryptographically signed digital lab report — one that cannot be altered after approval — for assessment and audit purposes?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q34. How manageable do you think it would be to review and digitally approve student submissions, given your current workload?

☐ Very manageable — I have adequate time ☐ Very challenging — would require additional staffing
☐ Manageable with minor workflow adjustments ☐ Not feasible given current workload
☐ Challenging but possible with support

Q35. Do you foresee any challenges in reviewing and approving digital practical submissions?

Q36. Would you prefer to approve submissions via a dedicated desktop portal, a mobile application, or both?

- ☐ Desktop portal only
- ☐ Mobile application only
- ☐ Both desktop and mobile
- ☐ No preference

SECTION G: Lab-Specific Questions

Please answer *ONLY* the subsection(s) that correspond to the laboratory/laboratories you manage. You may skip subsections that are not relevant to your role.

G1 — Physics Laboratories

Q37. What are the most common types of equipment used in your physics practicals? (Tick all that apply)

- ☐ Oscilloscopes
- ☐ Function generators
- ☐ Optical benches
- ☐ Vernier callipers / micrometers
- ☐ Power supplies
- ☐ Motion sensors / data loggers
- ☐ Other: _____

Q38. What specific challenges exist in recording and verifying experimental readings during physics practicals?

Q39. Would real-time digital recording of experimental readings improve accuracy and reduce transcription errors?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

G2 — Chemistry Laboratories

Q40. How are chemicals currently tracked and managed in your laboratory? (Select all that apply)

- ☐ Manual chemical inventory register
- ☐ Dedicated chemical management software
- ☐ Spreadsheet (Excel/Google Sheets)
- ☐ Informal tracking by technician memory
- ☐ Other: _____

Q41. What safety monitoring systems are currently in place in your chemistry laboratory?

- ☐ Gas leak detectors
- ☐ Automated fume hood monitoring
- ☐ Manual safety checks by technician
- ☐ Temperature/humidity sensors
- ☐ No formal monitoring system
- ☐ Other: _____

Q42. Would IoT-based real-time safety monitoring (e.g., gas, temperature, humidity sensors) improve laboratory safety outcomes?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

G3 — Engineering Laboratories

Q43. What types of equipment and machines are available in your engineering laboratory? (Tick all that apply)

- ☐ CNC machines / lathes
- ☐ Oscilloscopes / signal analysers
- ☐ 3D printers
- ☐ Welding equipment
- ☐ Electronic workstations
- ☐ Hydraulic/pneumatic rigs
- ☐ Other: _____

Q44. How is equipment usage currently scheduled for students?

- ☐ First-come, first-served on the day
- ☐ Advance booking via physical register
- ☐ Timetabled sessions managed by technician
- ☐ Online booking system
- ☐ No formal scheduling — informal arrangement

Q45. Would a digital equipment tracking and scheduling system improve utilisation rates and reduce student waiting times?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

G4 — Clinical / Medical Laboratories

Q46. How are patient samples currently tracked and documented in your laboratory?

Q47. How is chain-of-custody currently maintained for samples handled by students?

- ☐ Physical chain-of-custody forms signed at each transfer point
 ☐ No formal chain-of-custody system
☐ Electronic LIMS system
 ☐ Other: _____
☐ Informal verbal hand-offs

Q48. Would a digital chain-of-custody system integrated with USLE improve the reliability and auditability of sample handling?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q49. What is your greatest concern regarding student access to patient samples and clinical data?

SECTION H: System Flexibility and Inter-Lab Collaboration

Q50. How often do students need to redo or reschedule practical sessions at your laboratory?

- ☐ Rarely (a few per semester)
 ☐ Frequently (weekly)
☐ Occasionally (monthly)
 ☐ Very frequently (daily)

Q51. Would allowing students from other departments to access your laboratory (with appropriate authorisation) improve overall practical completion rates?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q52. How useful would inter-laboratory equipment sharing (e.g., Physics using Chemistry sensors) be for your institution?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q53. Do you support the idea of USLE being extended to support collaboration with external/private laboratories?

Note: Examples: industry research labs, hospital laboratories, private testing facilities.

- ☐ Yes
 ☐ No
 ☐ Not Sure

If yes, please explain:

Q54. What safeguards or conditions would you require before supporting external laboratory collaboration?

SECTION I: Implementation and Technology Adoption

Q55. How ready do you feel your laboratory is to adopt a digital management system like USLE?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q56. What type and level of training would you require before using the USLE system effectively?

- | | |
|---|--|
| ☐ Basic one-day orientation session | ☐ Self-paced online training materials |
| ☐ Multi-day hands-on workshop | ☐ Peer-to-peer training from early adopters |
| ☐ Ongoing support and refresher training | ☐ I would not require training |

Q57. What do you consider to be the most significant barriers to implementing USLE at JKUAT?

- | | |
|---|---|
| ☐ High initial cost of hardware (fingerprint scanners, RFID readers, etc.) | ☐ Privacy and data security concerns |
| ☐ Unreliable internet connectivity | ☐ Lack of institutional policy support |
| ☐ Resistance to change from staff or students | ☐ Other: _____ |
| ☐ Insufficient technical support for maintenance | |

Q58. What additional infrastructure would need to be in place before USLE could be successfully deployed?

Q59. How strongly do you agree that the benefits of USLE would outweigh the costs and effort of implementation?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

SECTION J: Final Feedback and Recommendations

Q60. What specific improvements or additional features would you suggest for the USLE system to better serve your laboratory?

Q61. Overall, do you believe the Unified Smart Laboratory Ecosystem would improve laboratory management and student outcomes at JKUAT?

- ☐ Yes
 ☐ No
 ☐ Not Sure

If yes, please explain:

Q62. How likely are you to personally adopt and champion the use of USLE in your laboratory if it were introduced?

1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
☐	☐	☐	☐	☐

Q63. Any additional comments, observations, or recommendations for the research team?

Thank you for your time and valuable input. Your responses will directly contribute to the design and development of the Unified Smart Laboratory Ecosystem at JKUAT.

For queries regarding this research, please contact the principal investigator through the Department of Information Technology.